

# Myeloid Neoplasms

Natasha M. Savage, MD, FCAP • designed to follow alongside the 44-slide lecture deck

Official Week 7 LOs

Slide-order companion

4 ScholarRx Bricks

Morphology + genetics

Faculty gap-fills labeled

Factoid sheet companion

## Overview: Official Week 7 Learning Objectives

**LO 1.** Distinguish acute from chronic myeloid neoplasms based on cellular differentiation, tempo, and typical clinical consequences.

**LO 2.** Recognize major AML categories conceptually, including blasts and key distinguishing features, and understand that genetic abnormalities influence prognosis and therapy without requiring detailed classification.

**LO 3.** Explain acute promyelocytic leukemia (APL): PML-RARA, DIC, and the rationale for differentiation therapy with ATRA.

**LO 4.** Have a basic understanding of myelodysplastic syndromes, including tempo, characteristic blood-count abnormalities, and typical clinical consequences.

**LO 5.** Describe CML as a prototype of targeted therapy, including BCR-ABL and constitutive tyrosine kinase signaling.

**LO 6.** Identify the core molecular drivers and key clinical features of PV, PMF, and ET, emphasizing JAK-STAT dysregulation, blood-count abnormalities, thrombosis, marrow fibrosis, and splenomegaly.

#### SCHOLARRX BRICKS PAIRED WITH THIS LECTURE

### Chronic Myeloproliferative Disorders: Foundations & Frameworks

Acute Myeloid Leukemia

Chronic Myeloid Leukemia

Polycythemia Vera

#### LO 1

**Slides 4-8:** myeloid lineage, blasts, acute vs chronic, AML/MDS/MPN/MDS-MPN framework.

#### LO 2-3

**Slides 9-16:** AML marrow failure, Auer rods/genetics, APL morphology, PML-RARA, DIC, ATRA, differentiation syndrome.

#### LO 4

**Slides 18-24:** ineffective hematopoiesis, cytopenias, dysplasia, pseudo-Pelger-Huet cells, ring sideroblasts, risk scoring.

#### LO 5

**Slides 29-34:** CML blood/marrow pattern, BCR-ABL1, Philadelphia chromosome, low LAP, imatinib and molecular monitoring.

#### LO 6

**Slides 35-39:** PV, ET, PMF - JAK2/CALR/MPL, dominant cell line, complications, and morphology.

#### SOURCE HIERARCHY

The **44-slide Savage deck** is the primary slide-order source. The uploaded “Lecture 4 and 5” companion says it was based on another 2026 version and adds treatment, CMML, and board-style distinctions. I use those additions only when they help the official objectives and label them as companion enrichment. The official LO sheet is the authority for what deserves emphasis.

THE ONE FRAMEWORK THAT MAKES THIS LECTURE MANAGEABLE

**AML = too many immature blasts.**

**MDS = cells are made badly and die / fail to exit marrow.**

**MPN = too many mature myeloid cells are produced.**

**MDS/MPN = dysplasia + proliferation together.**

Slides 4-8

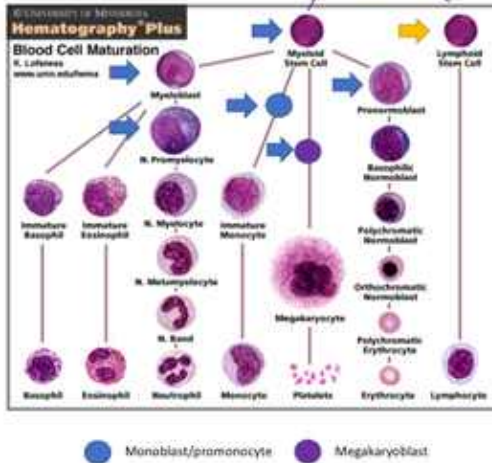
Definitions, lineage, and acute vs chronic disease

LO 1

#### Slide 4: define the words before memorizing diseases

- **Myeloid** includes granulocytes, monocytes, erythroid precursors, and megakaryocytes/platelets - everything hematopoietic outside the lymphoid lineage.
- **Acute leukemia** is blast-rich, rapidly progressive, and fatal without prompt treatment.
- **Chronic leukemia** contains predominantly more mature cells and usually has a more indolent course.
- A myeloid blast mass outside marrow/blood is called a **myeloid sarcoma**.

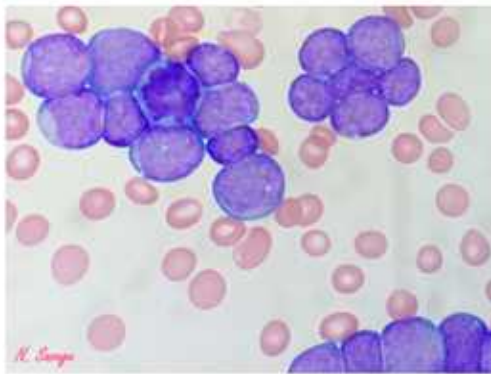
## Basics



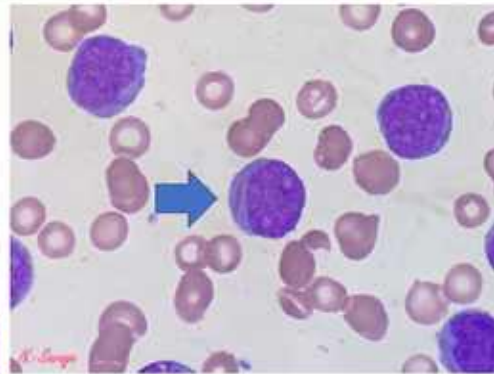
- A common stem cell makes the myeloid and lymphoid stem cells
- Myeloid stem cell goes on to make granulocytes (neutrophils, eosinophils, basophils), monocytes, RBC, megakaryocytes/platelets
- Arrows indicate a stage where malignant "blasts" can develop
  - Blue arrows pertain to AML

**Lecture slide 5 - full slide retained.** The blue/purple arrows show stages at which malignant myeloid blasts can arise. Use this to remember that AML is not only a neutrophil-precursor disease; monocytic, megakaryocytic, erythroid, and other myeloid blast phenotypes exist.

## Basics



Blasts are frequently large with high nuclear to cytoplasmic ratio, fine chromatin, scant to moderate cytoplasm, and sometimes prominent nucleoli.



Morphologic features can give you a clue to the origin of a blast, but the only definitive feature is an Auer rod which is comprised of abnormally fused primary granules (seen in early granulocyte precursors) indicating myeloid (not lymphoid) lineage.

**Lecture slide 6 - full slide retained.** Blasts have high nuclear-to-cytoplasmic ratio, fine chromatin, and sometimes nucleoli. Savage emphasizes that an Auer rod - fused primary granules - is definitive evidence of myeloid rather than lymphoid lineage.

### NEED TO MEMORIZE

**Auer rod = myeloid.**

It is essentially a needle-like bundle of abnormal primary granules and is highly useful when morphology is otherwise ambiguous.

### ACUTE VS CHRONIC IS ABOUT DIFFERENTIATION + TEMPO

The key difference is not simply "high WBC vs low WBC." Acute leukemia is dominated by **immature blasts that crowd out normal marrow**. Chronic disease is dominated by more mature malignant cells that may circulate in enormous numbers but retain more differentiation.

## LO 1 checkpoint

Can you define a blast, state what an Auer rod tells you, and explain why a chronic myeloid neoplasm can have a huge WBC yet still be “chronic”?

Slides 9-16

## AML and the APL emergency

LO 2 + LO 3

### Slide 9: AML in general

- Classic threshold: **20% or more myeloid blasts/blast equivalents** in blood and/or marrow.
- Savage notes important genetic exceptions in which the defining cytogenetic abnormality can establish AML below 20% blasts.
- Marrow crowding produces the clinical triad: **anemia -> fatigue/pallor; thrombocytopenia -> bleeding/bruising; neutropenia -> infection.**
- Modern WHO/ICC classification uses morphology + immunophenotype + cytogenetic/molecular information.

### OFFICIAL LO SAYS CONCEPTUAL - DO NOT DROWN IN CLASSIFICATION TABLES

The official objective explicitly says you should appreciate that genetics changes prognosis and therapy **without requiring detailed classification**. Savage also labels the detailed WHO/ICC MDS tables “No need to memorize these.” Learn the recurring genetic fingerprints, not every line of the classification.

| AML genetic clue       | Lecture association                          | What to remember                                                                 |
|------------------------|----------------------------------------------|----------------------------------------------------------------------------------|
| <b>t(8;21)</b>         | Subset of AML with maturation                | Generally favorable prognosis in the lecture.                                    |
| <b>inv(16)</b>         | Myelomonocytic AML with abnormal eosinophils | Harlequin eosinophils; favorable; monocytic disease can infiltrate gums/tissues. |
| <b>KMT2A/MLL 11q23</b> | Often monocytic differentiation              | Poor prognosis in the lecture; gingival/organ/tissue involvement can occur.      |
| <b>PML-RARA</b>        | APL                                          | <b>DIC emergency + differentiation therapy.</b>                                  |

## Acute promyelocytic leukemia (APL)

### Acute promyelocytic leukemia (APL) with *PML::RARA* fusion: pathology

We will focus on acute promyelocytic leukemia with *PML-RARA* (APL): 5-10% of AML cases

- Retinoic acid receptor gene on chromosome 17 is placed next to part of the *PML* gene on chromosome 15, creating a fusion gene that produces a chimeric protein (*PML-RARA*) which interferes with the differentiation of promyelocytes (stunts their growth; blocks their maturation)
- They have a distinct morphology as they look like unusual promyelocytes, frequently with abundant Auer rods
- They have a distinct immunophenotype: CD34/HLA-DR negative with strong MPO (Myeloperoxidase)
- They almost always present with some level of DIC (Disseminated Intravascular Coagulation) as the granules have procoagulants inappropriately turning on the coagulation cascade
  - Remember your DIC labs: PT/PTT, fibrinogen, d-dimer, CBC data, blood smear findings, reticulocytes, LDH, haptoglobin, bilirubin
  - Giving patients traditional chemotherapy would kill these cells, releasing their contents and exacerbating DIC potentially resulting in patient death
- In general, diagnosis is straightforward on morphology, flow cytometry (evaluates immunophenotype) helps support diagnosis, and FISH confirms revealing fusion of *PML* and *RARA* (can be done rapidly, but clinicians do not wait on result in general)



**Lecture slide 11 - full slide retained.** APL is a maturation block at the promyelocyte stage. The slide emphasizes abnormal promyelocytes with abundant Auer rods, strong MPO, CD34/HLA-DR negativity, and a very strong DIC association.

#### APL COLD SET

**t(15;17) / PML-RARA -> abnormal promyelocytes -> DIC -> start ATRA immediately when suspected.**

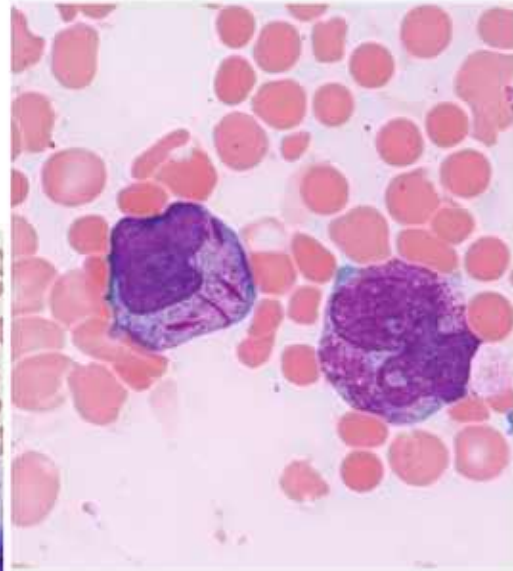
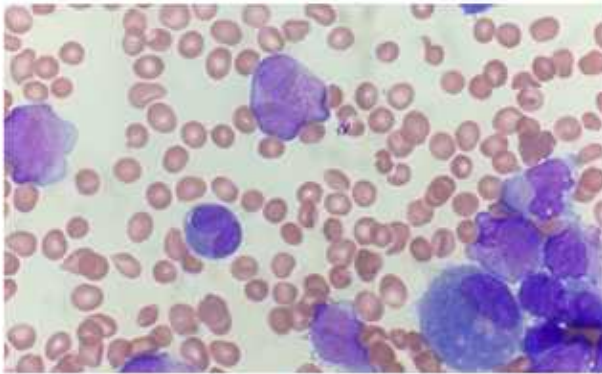
#### MECHANISM OF ATRA

PML-RARA blocks normal retinoic-acid receptor signaling and arrests differentiation. **ATRA binds the RARA portion and relieves/degrades the differentiation block**, allowing malignant promyelocytes to mature instead of simply being lysed en masse.



Acute promyelocytic leukemia (APL) with *PML::RARA* fusion; pathology

*For completeness' sake: there is a hypogranular variant that can be trickier to diagnose on morphology but still has DIC association (granules are there, they are just sub-microscopic)*



**Lecture slide 12 - full slide retained.** The hypogranular APL variant can look less obviously granulated on routine microscopy, but the DIC risk remains because the procoagulant granules are still present at a submicroscopic level.

#### WHY THE LECTURE SAYS TIME IS OF THE ESSENCE

Savage explicitly says to **start ATRA before confirmatory FISH returns** when APL is strongly suspected. The immediate cause of death is often hemorrhage/DIC, so waiting for perfect certainty is dangerous.



## Case 1

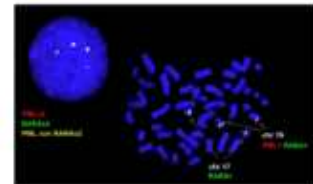
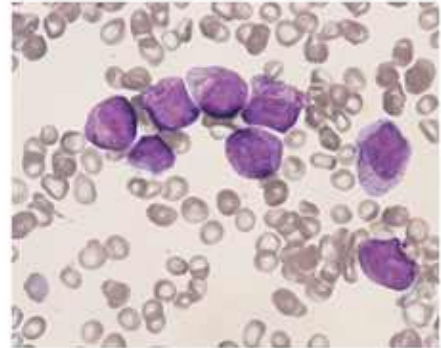
42 y/o male with past medical history of anxiety, who presents to the emergency department with hematuria

Initial assessment included CBC showing WBC 4K/L, Hgb 9 g/dL, plts 46K/L

PT/PTT/LDH elevated, haptoglobin/fibrinogen suppressed

Blood smear is shown in the picture

What is the next best step?



**Lecture slide 14 - full slide retained.** The case combines hematuria, thrombocytopenia, anemia, prolonged PT/PTT, low fibrinogen, and abnormal promyelocytes. The intended action is immediate ATRA while rapidly confirming PML-RARA.

### Slide 15: differentiation syndrome

- Occurs days to weeks after differentiation therapy as maturing myeloid cells release cytokines.
- Think **dyspnea/pulmonary edema, weight gain/edema, fever, hypotension, rising WBC, renal/liver abnormalities, pleural/pericardial effusions.**
- Lecture treatment: **dexamethasone**; severe cases may require holding ATRA/ATO.

## LO 2-3 checkpoint

Can you recognize AML from marrow-failure symptoms + blasts, then recognize APL specifically from abnormal promyelocytes/Auer rods + DIC and state why ATRA is started immediately?

Slides 18-24

## Myelodysplastic syndromes: ineffective hematopoiesis LO 4

### Slide 19: the paradox is the diagnosis

MDS is a clonal stem-cell disorder in which marrow cells are **dysplastic and ineffective**. The marrow can be hypercellular, yet the peripheral blood shows **cytopenias** because abnormal cells fail to mature/exit or die in the marrow.

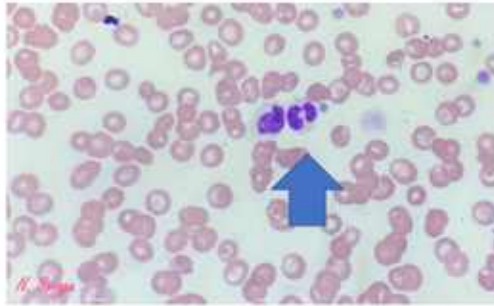
### MDS ONE-LINER

**Hypercellular dysfunctional marrow + peripheral cytopenias + dysplasia + risk of progression to AML.**

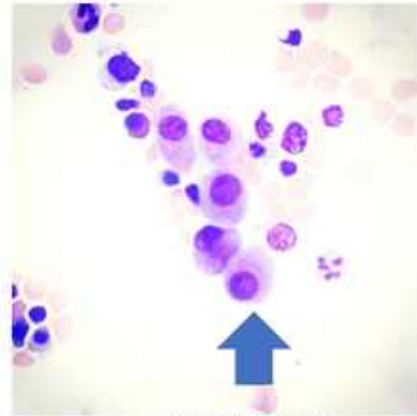
### Clinical logic

- More common in older adults.
- Can be found incidentally or present with anemia, infection, or bleeding from cytopenias.
- Blasts remain below the AML threshold; progression to AML reflects clonal evolution and increasing blasts/genetic complexity.
- Risk models in the deck: **IPSS-R** (counts, blasts, cytogenetics) and **IPSS-M** (adds molecular mutations).

## Myelodysplastic syndromes (MDS): pathology



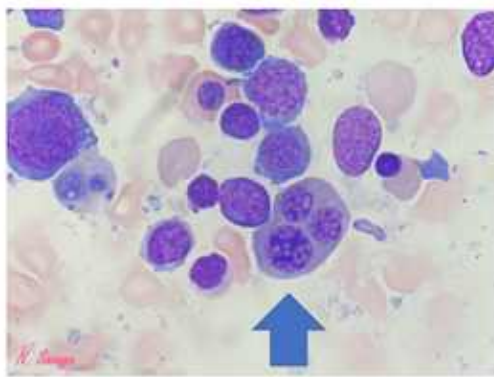
Hypogranular bi-lobed neutrophil; pseudo Pelger-Huet cell



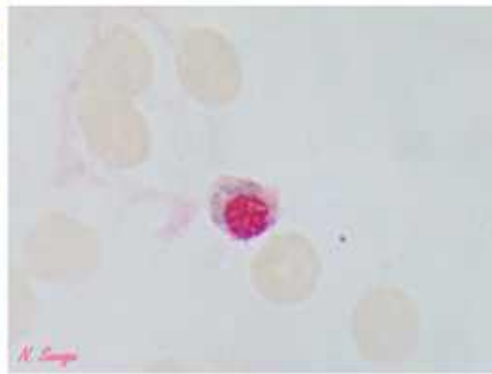
Micromegakaryocytes

**Lecture slide 22 - full slide retained.** Two morphology anchors: a hypogranular bilobed pseudo-Pelger-Huet neutrophil and abnormally small/dysplastic megakaryocytes.

## Myelodysplastic syndromes (MDS): pathology



Erythroid precursor with 3 nuclei



Ring sideroblast (seen on iron stain): indicative of abnormal iron incorporation into mitochondria around the RBC nucleus

**Lecture slide 23 - full slide retained.** The slide shows multinucleated erythroid dysplasia and a ring sideroblast on iron stain, where iron-loaded mitochondria encircle the erythroid nucleus.

### COMPANION TREATMENT ENRICHMENT

The companion adds a broad treatment map: supportive transfusion/infection care, hypomethylating agents such as azacitidine/decitabine for selected patients, lenalidomide in del(5q) disease, and allogeneic transplant as the potentially curative approach for suitable higher-risk patients. Treatment is risk-adapted rather than one-size-fits-all.

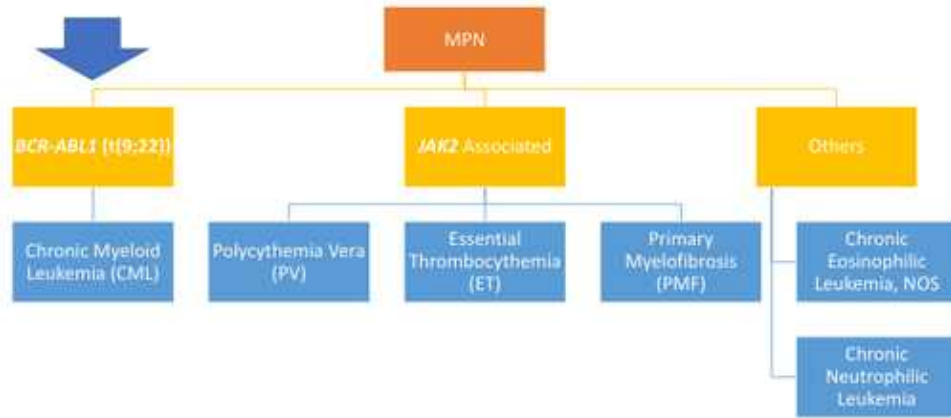
### LO 4 checkpoint

Can you explain how a patient can have a hypercellular marrow yet low circulating blood counts, and identify pseudo-Pelger-Huet cells, micromegakaryocytes, and ring sideroblasts as dysplasia clues?

**Slide 27: MPN is the opposite production problem from MDS**

- Clonal stem-cell mutations drive production of **too many maturing myeloid cells** that successfully enter blood.
- Often indolent and found in older adults.
- Splenomegaly is common because extramedullary hematopoiesis can develop.
- Any MPN can eventually transform to acute leukemia.

### Myeloproliferative neoplasms (MPN): in general



**Lecture slide 28 - full slide retained.** The deck separates BCR-ABL1-positive CML from the JAK2-associated MPNs PV, ET, and PMF. That split should organize the rest of the lecture.

## MPN FORK

**BCR-ABL1 positive -> CML.**

**BCR-ABL1 negative classic MPNs -> PV, ET, PMF, usually driven by JAK-STAT pathway mutations.**

Slides 29-34

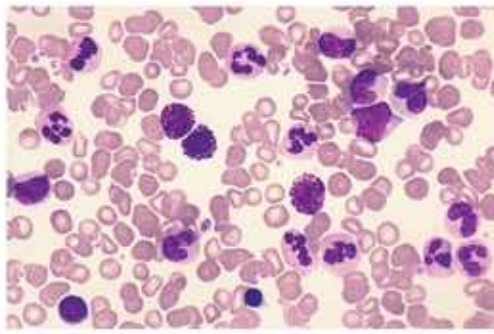
**CML: BCR-ABL1 and the prototype of targeted therapy**

LO 5

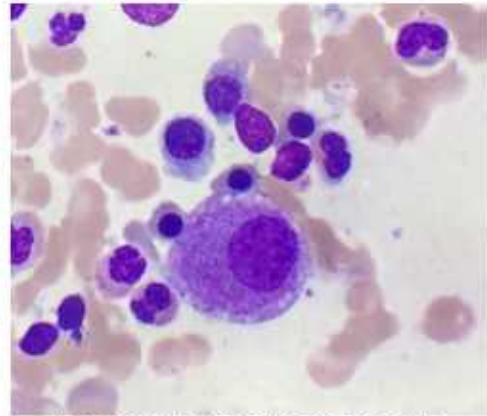
### Slides 29-32: recognize the blood picture

- Marked leukocytosis with a **full spectrum of granulocytic maturation / left shift**.
- **Basophilia** is especially important and helps separate CML from benign reactive leukocytosis.
- Marrow is hypercellular with granulocytic proliferation and characteristic small “dwarf” megakaryocytes.
- **LAP score is low** in CML, unlike many leukemoid reactions.

## Chronic myeloid leukemia (CML) with *BCR-ABL1*: pathology



Peripheral blood: notice leukocytosis with basophilia



Bone marrow: notice dwarf megakaryocyte

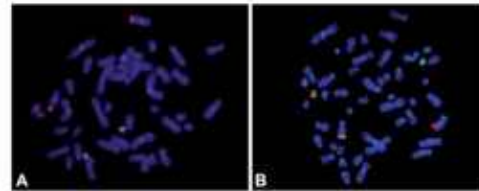
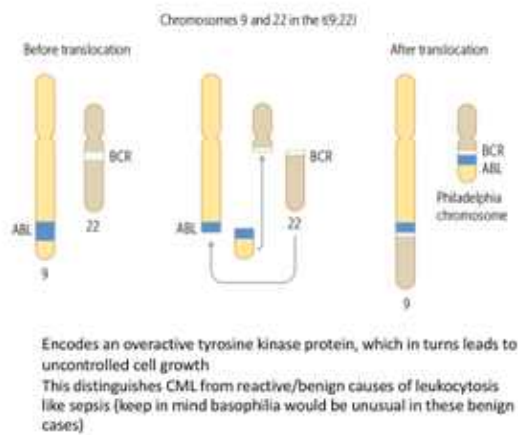
**Lecture slide 30 - full slide retained.** The blood image demonstrates granulocytic leukocytosis with basophilia; the marrow image shows a small monolobated dwarf megakaryocyte.

### CML COLD SET

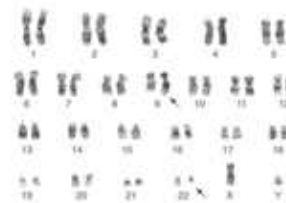
**t(9;22) Philadelphia chromosome -> BCR-ABL1 constitutive tyrosine kinase -> granulocytic leukocytosis + basophilia + low LAP + splenomegaly.**



## Chronic myeloid leukemia (CML) with *BCR-ABL1*: pathology



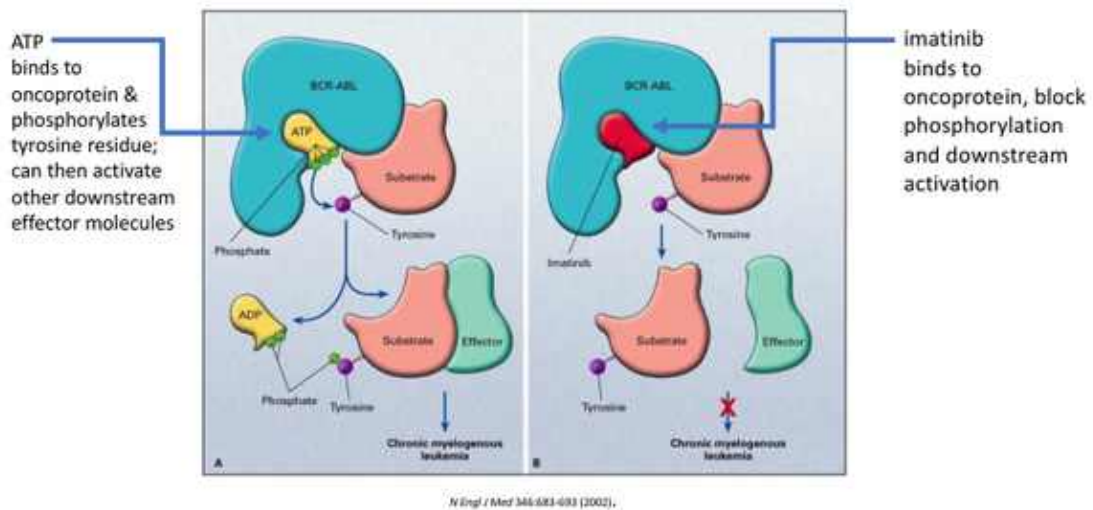
Can be detected by FISH rapidly



Also seen on karyotype  
Can be monitored via PCR

**Lecture slide 32 - full slide retained.** BCR-ABL1 can be detected by cytogenetics/FISH and monitored quantitatively by PCR. The molecular lesion explains both diagnosis and response monitoring.

## CML therapy: Focused on Inhibition of *BCR-ABL* Oncoprotein



**Lecture slide 33 - full slide retained.** The lecture's targeted-therapy diagram: ATP binding normally permits phosphorylation and downstream signaling; imatinib occupies the kinase and blocks that signaling.

### TARGETED THERAPY MEMORY BOX

Ozzy study break - click him for encouragement.

**CML changed oncology because the lesion is both diagnostic and druggable:** BCR-ABL1 creates the kinase; a TKI such as imatinib directly blocks it.

### COMPANION BOARD ENRICHMENT - PHASES

The companion gives the classic progression model: **chronic phase <10% blasts -> accelerated phase 10-19% -> blast phase 20% or more**, with blast crisis behaving as acute leukemia. The current 44-slide deck focuses more on the chronic-phase phenotype and targeted-therapy concept.

#### CML VS LEUKEMOID REACTION

Both can produce high neutrophils/left shift. **CML favors basophilia, low LAP, and BCR-ABL1.** A reactive infection-associated leukemoid response generally has high LAP and lacks BCR-ABL1.

#### LO 5 checkpoint

Can you go from t(9;22) -> BCR-ABL1 kinase -> CML blood pattern -> imatinib mechanism, and explain why PCR is useful after diagnosis?

Slides 35-39

JAK-STAT MPNs: PV, ET, and PMF

LO 6

## Polycythemia vera (PV)

### JAK2 associated MPN: PV pathology

- Predominated by RBC proliferation
- Blood is thicker than normal (too many RBC)
- Patients may develop thromboses, MI, stroke, DVT, PE, headache, vision issues, etc.
  - May also develop hemorrhages
- Blood may show thrombocytosis and leukocytosis, but polycythemia is main feature
- Bone marrow is hypercellular with panmyelosis
- Essentially all cases associated with a JAK2 activating mutation resulting in constitutive activation of a tyrosine kinase involved in cell proliferation
- Often asymptomatic at presentation with an indolent course
  - Itching (often after a hot shower) may be a unique symptom
    - Mast cell degranulation leads to histamine release
    - Another hypothesis: RBC release ADP when the skin cools down and this somehow leads to platelets aggregation in blood vessels, causing itching
  - Face may show a reddish discoloration called plethora that is presumably caused by stasis in the vessels of the face
  - Erythromelalgia (burning pain in the hands or feet) due to microvascular thrombi; respond well to aspirin
- As time goes on and disease progresses, fibrosis may develop leading to cytopenias and worsening splenomegaly
- Just like CML must be distinguished from benign leukocytosis, PV must be distinguished from benign polycythemia (check JAK2)
  - Secondary polycythemia: hypoxia, high altitude, smoking, obesity, sleep apnea, certain tumors, hemoglobinopathies (check EPO level)
  - Spurious polycythemia: dehydration



**Lecture slide 36 - full slide retained.** PV is dominated by RBC proliferation and panmyelosis. The slide emphasizes thrombosis, aquagenic pruritus, plethora, erythromelalgia, and the need to distinguish true PV from secondary/spurious erythrocytosis.

#### PV RECOGNITION SET

**High Hgb/Hct + JAK2 mutation + low EPO + aquagenic pruritus / erythromelalgia / thrombosis.**

#### WHY EPO IS LOW

PV is autonomous marrow signaling, so high RBC mass suppresses renal EPO through normal feedback. **Secondary erythrocytosis is driven by EPO and therefore tends to have an elevated/inappropriately normal EPO level.**

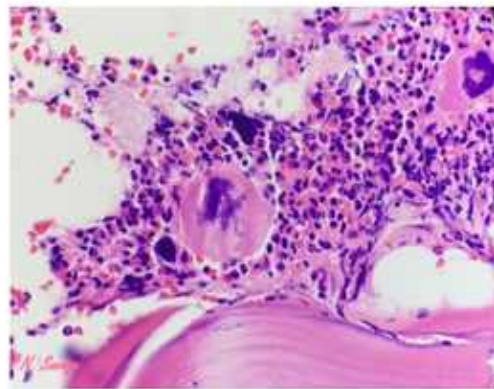
## COMPANION TREATMENT ENRICHMENT

Recognition-level treatment logic: phlebotomy lowers hematocrit; low-dose aspirin is commonly used for microvascular/thrombotic risk when appropriate; higher-risk patients may require cytoreduction such as hydroxyurea or JAK-pathway therapy.

## Essential thrombocythemia (ET)

### JAK2 associated MPN: ET pathology

- Usually, adults but seen in your patients too (especially young women)
- Predominated by thrombocytosis; significant leukocytosis and polycythemia are not expected
- As only the megakaryocytes are proliferating, bone marrow is normocellular to slightly hypercellular with large "staghorn" like megakaryocytes
- Development of marrow fibrosis is not typical
- 50% will have JAK2 mutations; those that don't may have CALR or MPL mutations
- Often asymptomatic but may present with excess bleeding or clotting
  - May also have erythromelalgia
- MOST patients with thrombocytosis do not have ET
  - Iron deficiency, malignancy, trauma, surgery, infection, acute blood loss, hyposplenism, etc.
- Typically, indolent course that may not alter lifespan



**Lecture slide 37 - full slide retained.** ET is dominated by thrombocytosis with large hyperlobulated 'staghorn' megakaryocytes; about half have JAK2, while CALR and MPL account for many JAK2-negative cases.

## ET RECOGNITION SET

**Persistent thrombocytosis + large mature hyperlobulated megakaryocytes + JAK2/CALR/MPL.**

The clinical paradox is **thrombosis OR bleeding.**

#### BOARD ENRICHMENT - WHY VERY HIGH PLATELETS CAN BLEED

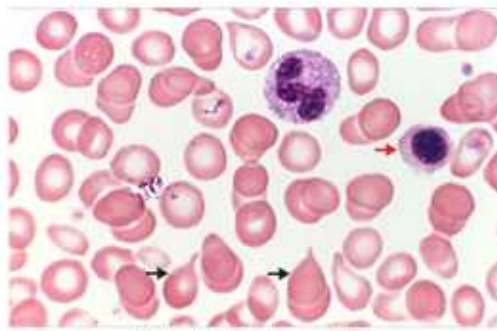
Extreme thrombocytosis can remove high-molecular-weight vWF multimers from circulation and produce **acquired von Willebrand disease**. So “more platelets” does not always mean “more clotting.”

### Primary myelofibrosis (PMF)

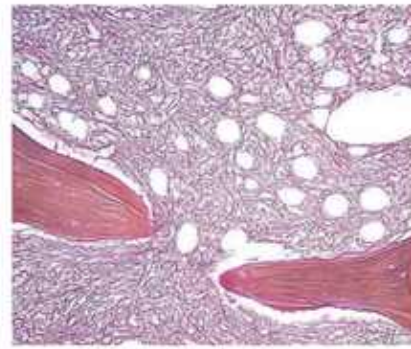
#### Slide 38: fibrosis is reactive to the neoplastic clone

- The malignant clone is especially megakaryocytic/granulocytic, but marrow fibrosis develops from **fibroblast-stimulating cytokines released by abnormal megakaryocytes**.
- Progressive fibrosis -> cytopenias, extramedullary hematopoiesis, and worsening splenomegaly.
- Common driver groups: **JAK2, CALR, MPL**.

## JAK2 associated MPN: PMF pathology



Myelophthitic blood smear: nRBC, immature granulocytes (not pictured), and teardrop cells (dacrocytes)



Reticulin stain on the bone marrow biopsy shows marrow fibrosis which may lead to a dry tap

**Lecture slide 39 - full slide retained.** The peripheral smear is myelophthitic with nucleated RBCs, immature granulocytes, and teardrop cells. Reticulin staining shows marrow fibrosis that can cause a dry tap.

### PMF RECOGNITION SET

**Massive splenomegaly + teardrop cells + leukoerythroblastic smear + dry tap + marrow fibrosis + JAK2/CALR/MPL.**

|                      | PV                                      | ET                                         | PMF                                                   |
|----------------------|-----------------------------------------|--------------------------------------------|-------------------------------------------------------|
| <b>Dominant cell</b> | RBC / panmyelosis                       | Platelets / megakaryocytes                 | Abnormal megakaryocytes + granulocytes, then fibrosis |
| <b>Drivers</b>       | JAK2 essentially all cases in lecture   | JAK2, CALR, MPL                            | JAK2, CALR, MPL                                       |
| <b>Signature</b>     | Low EPO, aquagenic pruritus, thrombosis | Marked thrombocytosis, thrombosis/bleeding | Teardrops, dry tap, massive spleen                    |
| <b>Marrow</b>        | Panmyelosis                             | Large hyperlobulated megakaryocytes        | Clustered atypical megakaryocytes -> fibrosis         |

## LO 6 checkpoint

Can you identify PV, ET, and PMF from which cell line dominates, their driver mutations, and their signature complication/morphology?

Slides 41-42

**MDS/MPN overlap and CMML - for completeness**

Beyond official core LO

## Slide 42: overlap means dysplasia + proliferation

Savage explicitly includes MDS/MPN overlap “for completeness” and names **chronic myelomonocytic leukemia (CMML)** as the most common example. It is useful as a framework but is not one of the six official Week 7 Myeloid objectives.



#### COMPANION GAP-FILL - CMML RECOGNITION

- Persistent absolute monocytosis **at least  $0.5 \times 10^9/L$  and at least 10% of WBCs**.
- Blasts **<20%**; 20% or more moves into AML territory.
- **No BCR-ABL1** (helps exclude CML).
- Dysplasia in at least one lineage supports the diagnosis.
- Companion separates proliferative vs dysplastic types around a WBC threshold of  $13 \times 10^9/L$ .

#### LOW-PRIORITY BUT USEFUL

**CMML = persistent monocytosis + dysplasia + proliferation + no BCR-ABL1 + blasts below AML threshold.**

Synthesis

One-table myeloid fingerprint map

LO 1-6

| Disease     | Core problem / genetics                                    | Blood or morphology clue                                                            | Clinical / treatment anchor                         |
|-------------|------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>AML</b>  | Myeloid blasts; often $\geq 20\%$ unless defining genetics | Blasts; Auer rod proves myeloid lineage                                             | Marrow failure; genetics direct risk/therapy        |
| <b>APL</b>  | PML-RARA t(15;17)                                          | Abnormal promyelocytes, many Auer rods                                              | <b>DIC emergency; ATRA immediately</b>              |
| <b>MDS</b>  | Ineffective dysplastic hematopoiesis                       | Cytopenias + hypercellular dysplastic marrow; pseudo-Pelger-Huet / ring sideroblast | Older adults; AML progression risk                  |
| <b>CML</b>  | BCR-ABL1 t(9;22) kinase                                    | Huge granulocytic left shift, basophilia, low LAP                                   | TKI prototype; PCR monitoring                       |
| <b>PV</b>   | JAK2-driven erythrocytosis                                 | High Hgb/Hct, panmyelosis, low EPO                                                  | Aquagenic pruritus, thrombosis, erythromelalgia     |
| <b>ET</b>   | JAK2/CALR/MPL megakaryocytic proliferation                 | Very high platelets, large hyperlobulated megakaryocytes                            | Thrombosis or bleeding; acquired vWD if extreme     |
| <b>PMF</b>  | JAK2/CALR/MPL clone -> cytokine-driven fibrosis            | Teardrops, leukoerythroblastosis, dry tap                                           | Massive splenomegaly / extramedullary hematopoiesis |
| <b>CMML</b> | MDS/MPN overlap                                            | Persistent monocytosis + dysplasia                                                  | Blasts $< 20\%$ , BCR-ABL1 absent                   |

#### MUTATION MAP TO MEMORIZE

**APL = PML-RARA (15;17).**

**CML = BCR-ABL1 (9;22).**

**PV = JAK2.**

**ET / PMF = JAK2, CALR, or MPL.**

#### COMMON EXAM TRAPS

- **Very high WBC does not automatically mean AML.** CML can have WBC >100K while cells are mostly maturing granulocytes.
- **Basophilia strongly favors CML** over a simple leukemoid response.
- **Low EPO favors PV**; high EPO favors secondary erythrocytosis.
- **Teardrops + dry tap** point toward marrow fibrosis, not ET.
- **APL is an emergency before FISH confirmation.**
- **MDS has cytopenias despite a busy marrow.**

## Learning-Objective Synthesis

| Official LO | What you should be able to do without notes                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| LO 1        | Distinguish acute blast-rich disease from chronic mature-cell disease and organize AML vs MDS vs MPN vs overlap.                           |
| LO 2        | Recognize an AML blast/Auer rod, explain marrow failure, and know why recurrent genetics matter without memorizing every WHO/ICC category. |
| LO 3        | State PML-RARA, DIC risk, ATRA mechanism, urgency, and differentiation syndrome.                                                           |
| LO 4        | Explain MDS ineffective hematopoiesis and recognize characteristic dysplasia/cytopenias and AML-progression risk.                          |
| LO 5        | Go from Philadelphia chromosome to BCR-ABL1 kinase to CML phenotype to imatinib/TKI targeted therapy and PCR monitoring.                   |
| LO 6        | Separate PV, ET, and PMF by dominant cell line, JAK-STAT driver, blood/marrow findings, and major complications.                           |

#### LAST 60 SECONDS

**AML:** blasts / Auer rod / marrow failure.

**APL:** 15;17 PML-RARA + DIC -> ATRA now.

**MDS:** hypercellular dysfunctional marrow + cytopenias + dysplasia.

**CML:** 9;22 BCR-ABL1 + basophilia + low LAP -> TKI.

**PV:** JAK2 + high Hct + low EPO + itching/thrombosis.

**ET:** JAK2/CALR/MPL + platelets + thrombosis/bleeding.

**PMF:** JAK2/CALR/MPL + teardrops + dry tap + massive spleen.

**Source hierarchy:** Savage's 44-slide Myeloid Neoplasms deck is the primary slide-order source. The official Week 7 objectives control emphasis. The uploaded Lecture 4 and 5 companion is used as supplemental explanation for broad treatment logic, CML phases, CMML, and classic board distinctions. Detailed classification tables explicitly marked "No need to memorize" are not turned into artificial memorization burden.